

PRATHAMESH MAHAVEER PATIL

AI/ML Engineer

☎ [9187585706](tel:9187585706) | ✉ prathameshece2022@gmail.com |  [LinkedIn](#) |  [GitHub](#)

PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building and deploying production-grade machine learning and GenAI systems, including RAG pipelines, recommendation engines, and real-time inference APIs. Experienced in vector databases, LLM integration, and workflow automation, with a strong focus on low-latency, scalable architectures and business impact.

PROFESSIONAL EXPERIENCE

AI/ML Engineer (GenAI & Recommendation Systems)

6D Technologies Ltd | Bengaluru, Karnataka | www.6dtechnologies.com

January 2023 — Present

- Built and deployed customer segmentation pipelines using clustering and RFM features, integrated into CRM systems for real-time campaign targeting.
- Designed and deployed hybrid recommendation system (collaborative filtering + XGBoost) with feature engineering and real-time inference APIs, improving conversion rates by 12–18%.
- Designed and deployed end-to-end RAG pipeline (data ingestion → embedding → vector search → LLM inference) using vector databases (Chroma/Redis) and LLM APIs, improving chatbot accuracy by 30–35%.
- Improved LLM reliability by optimizing retrieval quality, context filtering, and prompt constraints to reduce hallucinations.
- Optimized real-time inference latency by 40% (<300ms) through caching, efficient retrieval strategies, and optimized API design.
- Developed **Text-to-Rule system** converting natural language into executable business logic, reducing rule creation time by ~60%
- Developed event-driven ML/LLM workflows using N8N with webhook-based triggers for real-time automation and orchestration.
- Designed and deployed **real-time ML systems using FastAPI/REST APIs and webhook-based triggers**

- Worked on production-grade ML systems handling real-time inference and scalable AI workloads.
- Orchestrated ML pipelines using **Seahorse Workflow (ML pipeline orchestration similar to Airflow/MLflow)**
- Collaborated with cross-functional teams (7+ developers) to deliver scalable AI-driven solutions

KEY ACHIEVEMENTS

- Improved chatbot accuracy by 30–35% using RAG
- Reduced latency by 40% (<300ms)
- Increased recommendation conversion by 12–18%

TECHNICAL SKILLS

- **AI/ML & Generative AI**
RAG Pipelines, LLM Integration, Prompt Engineering, Embeddings, Vector Search, LangChain, LlamaIndex
- **Machine Learning**
XGBoost, Scikit-learn, Recommendation Systems (Collaborative + Hybrid), Customer Segmentation (Clustering, RFM), NLP
- **Backend & Model Serving**
FastAPI, REST APIs, Real-time Inference Systems, Streamlit
- **Data & Vector Databases**
Redis (Vector DB), Chroma, MySQL, SingleStoreDB, Supabase
- **Workflow & MLOps**
N8N (Event-driven automation, Webhooks), Apache Airflow, Seahorse Workflow
- **Programming & Tools**
Python, SQL, Git, Linux

KEY PROJECTS

AI-Powered Customer Segmentation System

- Applied unsupervised learning (clustering + RFM analysis) for customer segmentation
- Processed large-scale telecom customer data for behavioral insights
- Integrated outputs into CRM systems for targeted marketing campaigns

Product Recommendation Engine

- Developed **hybrid recommendation system** combining collaborative filtering and XGBoost
- Designed separate pipelines for prepaid and postpaid customer behavior
- Served predictions via **low-latency APIs (<300ms)** for real-time applications

RAG-Based Customer Value Management Chatbot

- Built end-to-end retrieval + generation pipeline using embeddings and vector search (Chroma, Redis)
- Integrated LLMs for contextual response generation with improved semantic understanding
- Reduced support workload by ~20% and improved response accuracy by ~30%+
- Deployed real-time chatbot with API + web UI integration

Text-to-Rule Automation System

- Built NLP-based system to convert natural language into executable business rules
- Implemented intent classification and entity extraction
- Reduced manual configuration effort by ~60%

ML Workflow & Automation Pipelines

- Built automated workflows for model training and validation using tools such as **Seahorse Workflow**, **N8N**, and **Apache Airflow** to streamline ML lifecycle management.
- Integrated data from multiple databases into scalable ML pipelines, ensuring efficient data preprocessing and model experimentation.
- Designed event-driven workflows and webhook-based triggers using N8N for real-time inference and data transformation.
- Utilized **Metabase** for monitoring data metrics, validating model outputs, and enabling business-level visibility into analytics workflows.

Interactive ML Dashboard (Streamlit)

- Developed web-based Streamlit applications for real-time model experimentation and business analytics.
- Created dashboards for customer insights, model performance monitoring, and prediction visualization.

- Built iframe-embedded chatbot UI components for seamless cross-platform integration.

PRODUCTION & SYSTEM DESIGN HIGHLIGHTS

- Built scalable ML systems with FastAPI-based REST APIs for real-time inference
- Designed low-latency RAG architecture using vector databases and optimized retrieval
- Implemented caching strategies to reduce inference cost and latency
- Developed event-driven pipelines using N8N and Airflow
- Monitored model performance using dashboards and logging systems

EDUCATION

Bachelor of Engineering (B.E.)

The Oxford College of Engineering | Bengaluru, Karnataka

Graduated. 2022 CGPA: 7.91/10

LANGUAGES

English, Kannada, Marathi, Hindi

PROFESSIONAL ATTRIBUTES

- Strong problem-solving skills with focus on delivering business value through AI/ML solutions
 - Experience working in Agile development environments with cross-functional teams
 - Excellent communication skills enabling effective collaboration with technical and non-technical stakeholders
 - Continuous learner staying updated with latest AI/ML trends and technologies
-